CAN WE PREDICT DEPRESSION RISK AMONG STUDENTS?

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Problem : Depression affects millions of students, yet it often goes unnoticed.

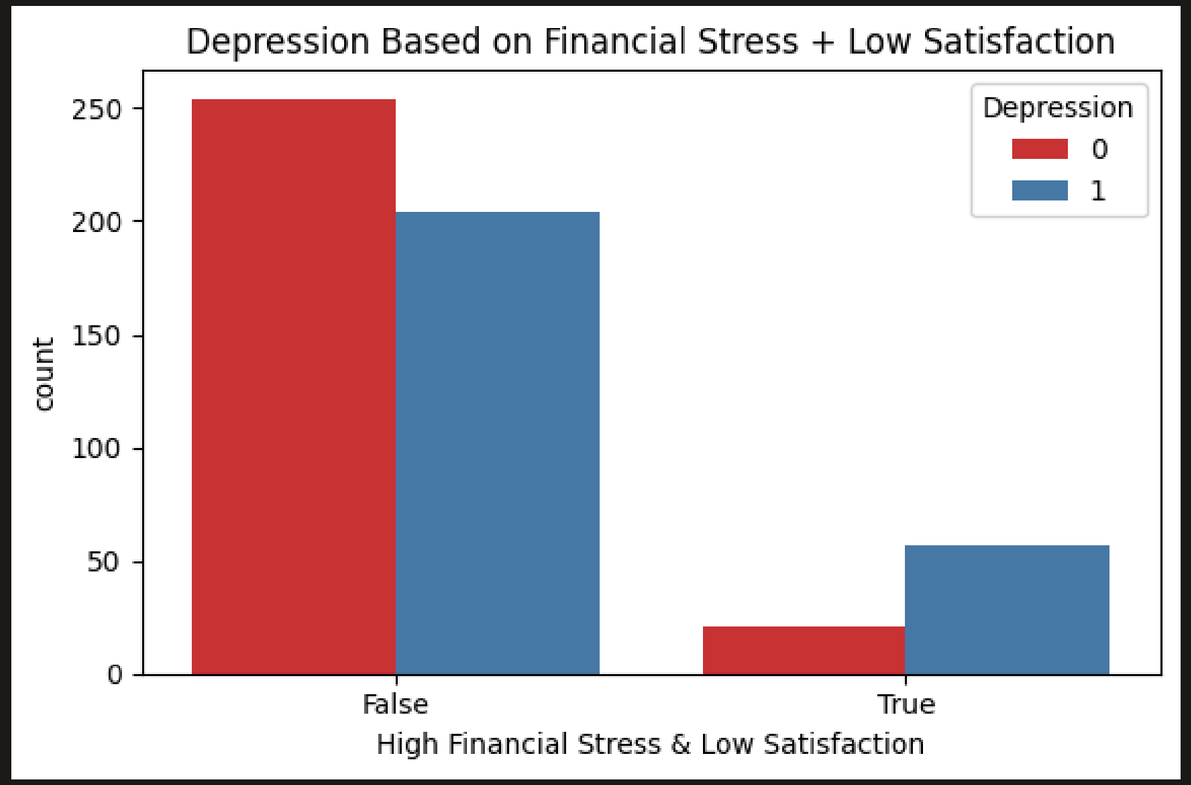
Goal: Can we predict depression risk using behavioral data and real online sentiment?

Data Sources:
- Student survey (study habits, pressure, satisfaction, etc.)
- Reddit posts (sentiment & topics around mental health)

Challenge : Build a model that detects hidden risk — not just patterns in the data.

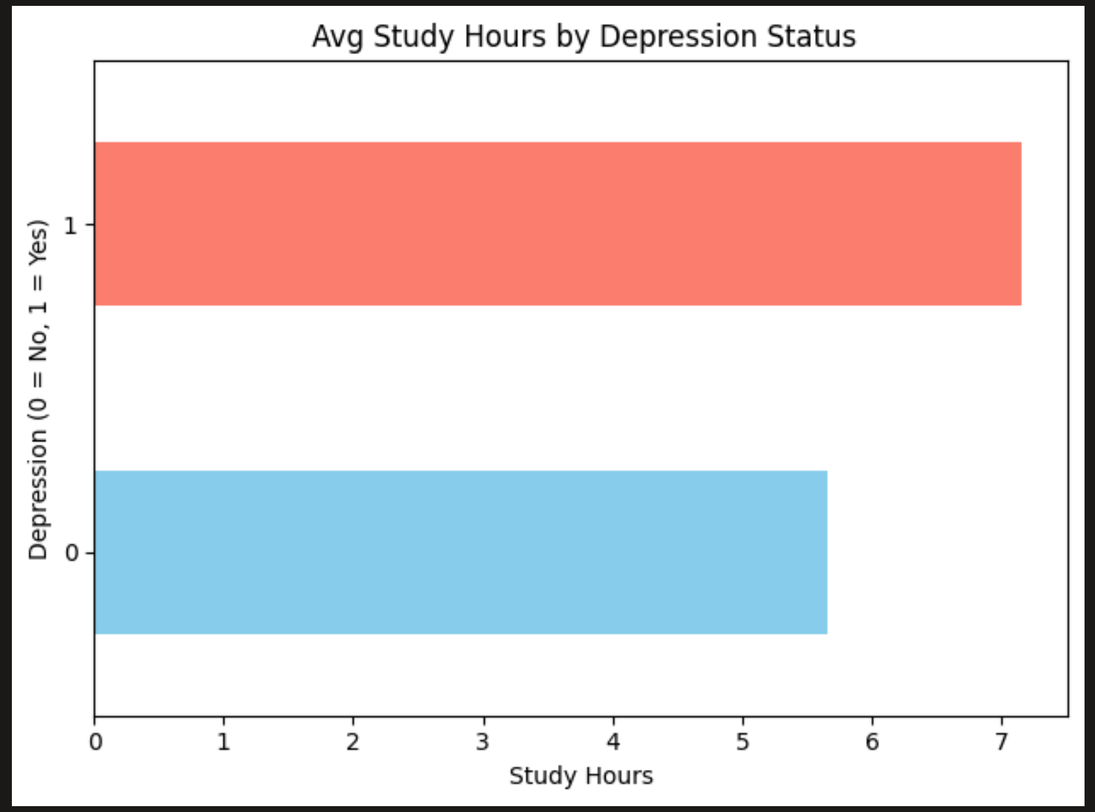
Academic Strain: Hidden Signals of Depression

Beyond the Obvious



Students with both high stress and low satisfaction show a sharp rise in depression cases.

When Expectations Break

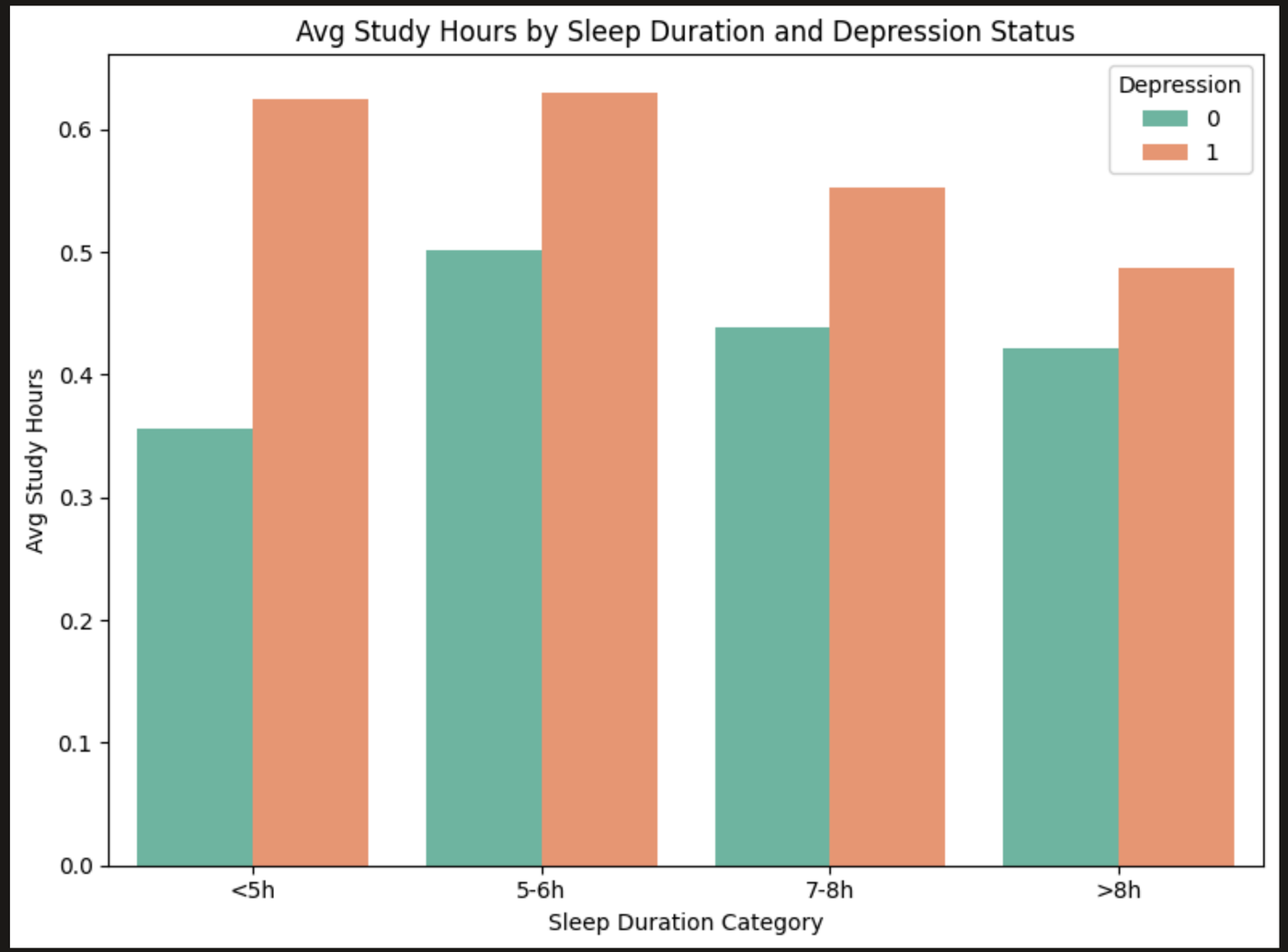


Surprisingly, depressed students study more, hinting at pressure or perfectionism.

Continuing the story of academic strain…

The Burnout Loop: When Studying More Fails to Help

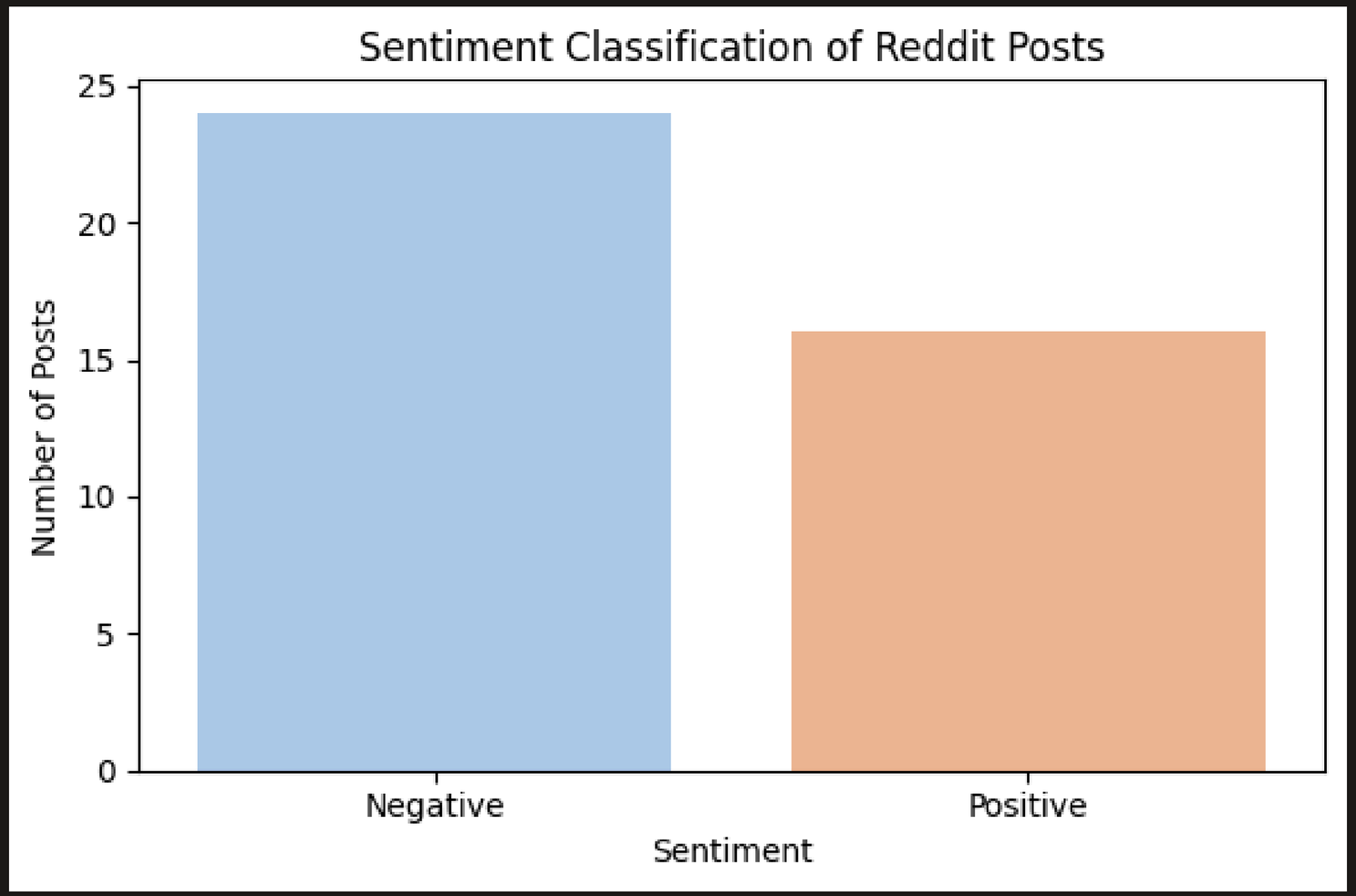
Less Sleep → More Studying → Still High Depression Risk



Students sleeping less than 6 hours tend to compensate by studying more, yet still experience higher depression risk. This highlights that overwork cannot replace rest for mental well-being

Raw & Real: What Students Are Actually Saying

1. **“Well this sucks”** — So second year of uni and I fucking hate it. No motivation, just pressure and loans.
2. **“I’m 23 and wanting to switch degrees — I feel stuck”** — I am currently in my final year and feel like everything is a mistake.
3. **“Senior and hate my major”** — I’m currently regretting all my life choices. This degree feels pointless.
4. **“Considering dropping Uni to Join College”** — I go to a prestigious university, but I feel burned out and lost.
5. **“Me and my anxiety before university exams”** — Waking up with panic attacks and zero sleep. University exams are too much for me.



To validate our findings, we turned to Reddit — where students post raw and unfiltered thoughts. Despite not being asked, most messages reflect stress, burnout, and loss of motivation.

Over 60% of Reddit posts carry a negative emotional tone — without being directly asked.

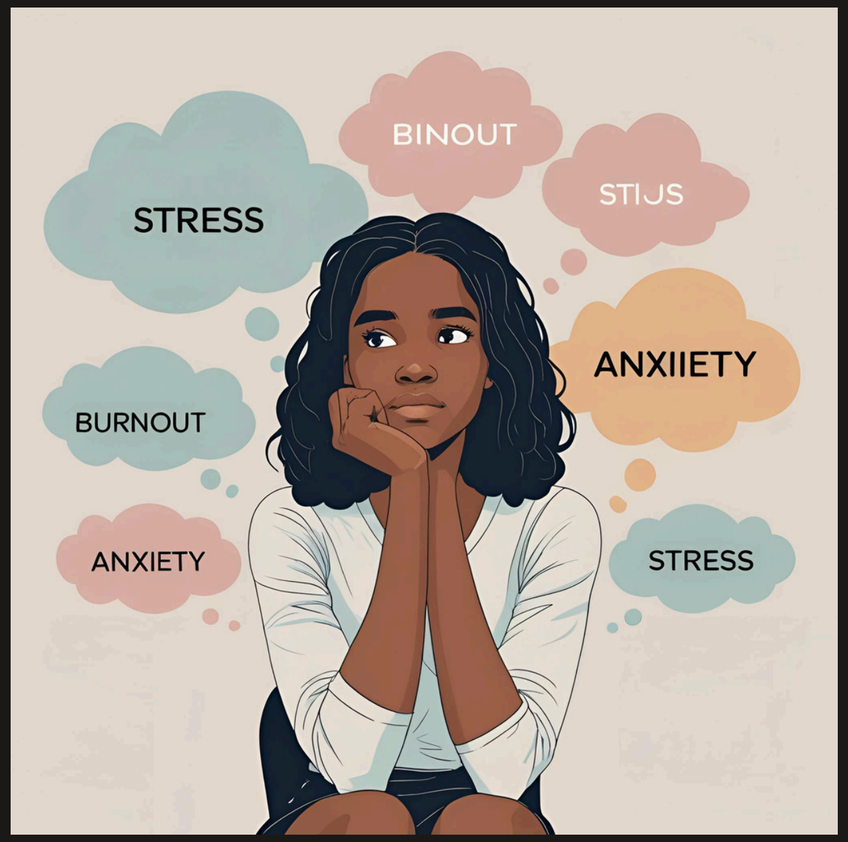
What Are Students Talking About? Top Themes Behind Negative Sentiment

Main Themes Identified:

Burnout & Exams
Keywords: exams, semester, burnout, study, degree

Performance Anxiety
Keywords: anxiety, presentation, students, lack, work

Doubts About University
Keywords: university, life, wanted, think, school



So what?

This analysis confirms that stress, anxiety, and doubt are common experiences among students.

But Can We Predict Who is at Risk?

So far, we saw clear patterns in survey data and online sentiment. Next, we wanted to test whether these signals can reliably predict depression risk among students.

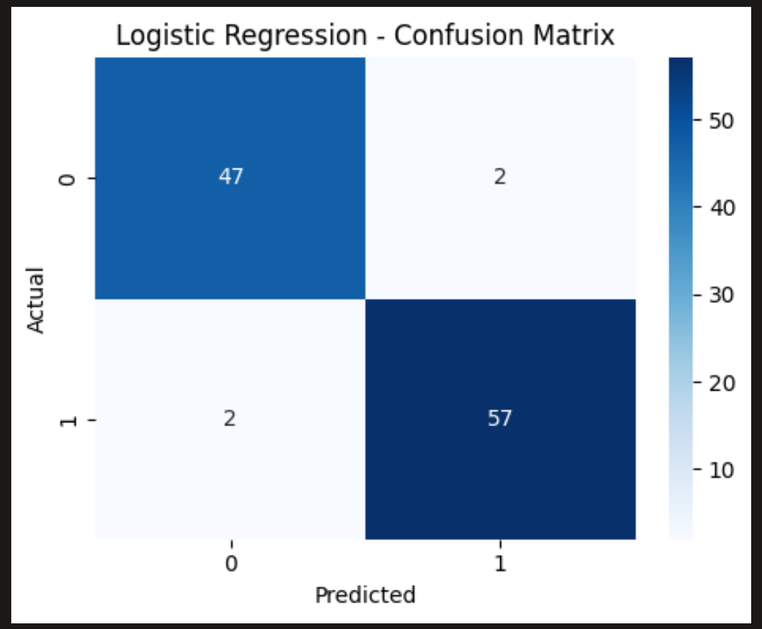
Predictive Models Used

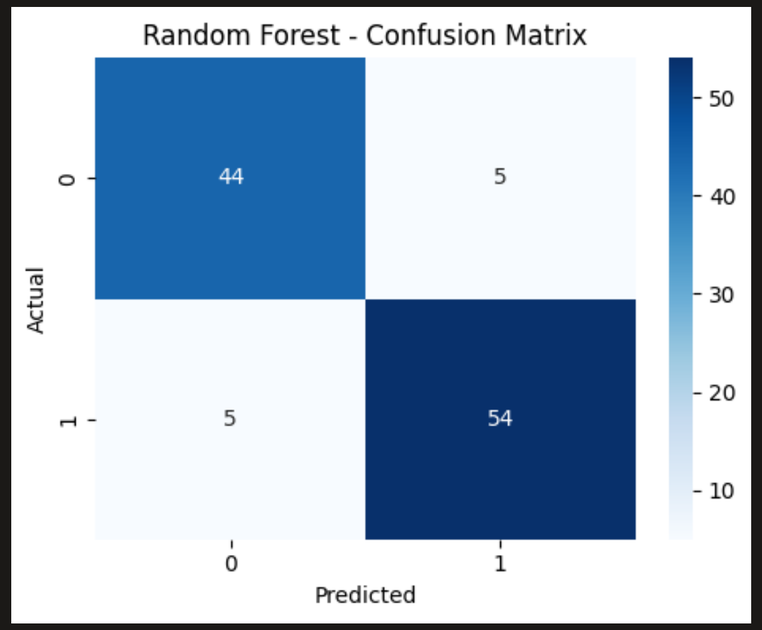
- Logistic Regression: interpretable baseline classifier
- Random Forest: robust ensemble model capturing non-linearities
- Goal: validate which factors truly matter for prediction

Visual comparison of prediction accuracy for each approach.

How Well Can We Predict?

Model	Accuracy
Logistic Regression	96%
Random Forest	91%



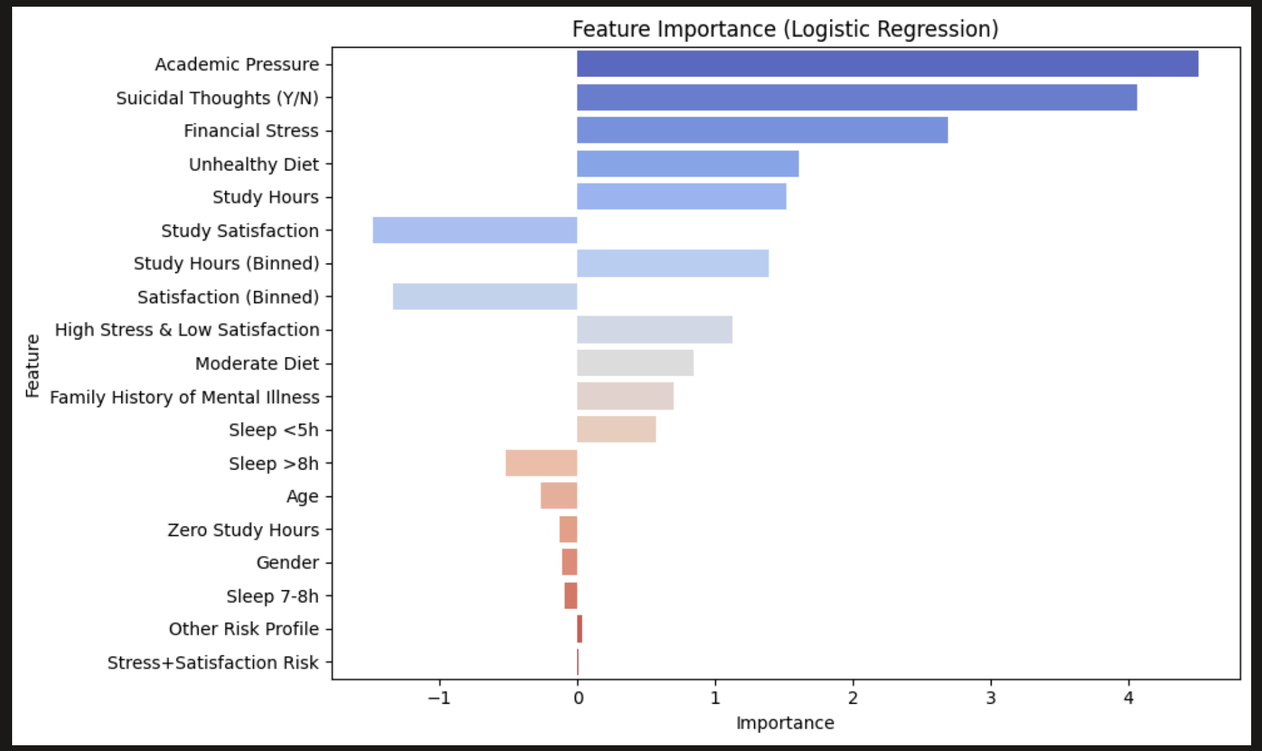


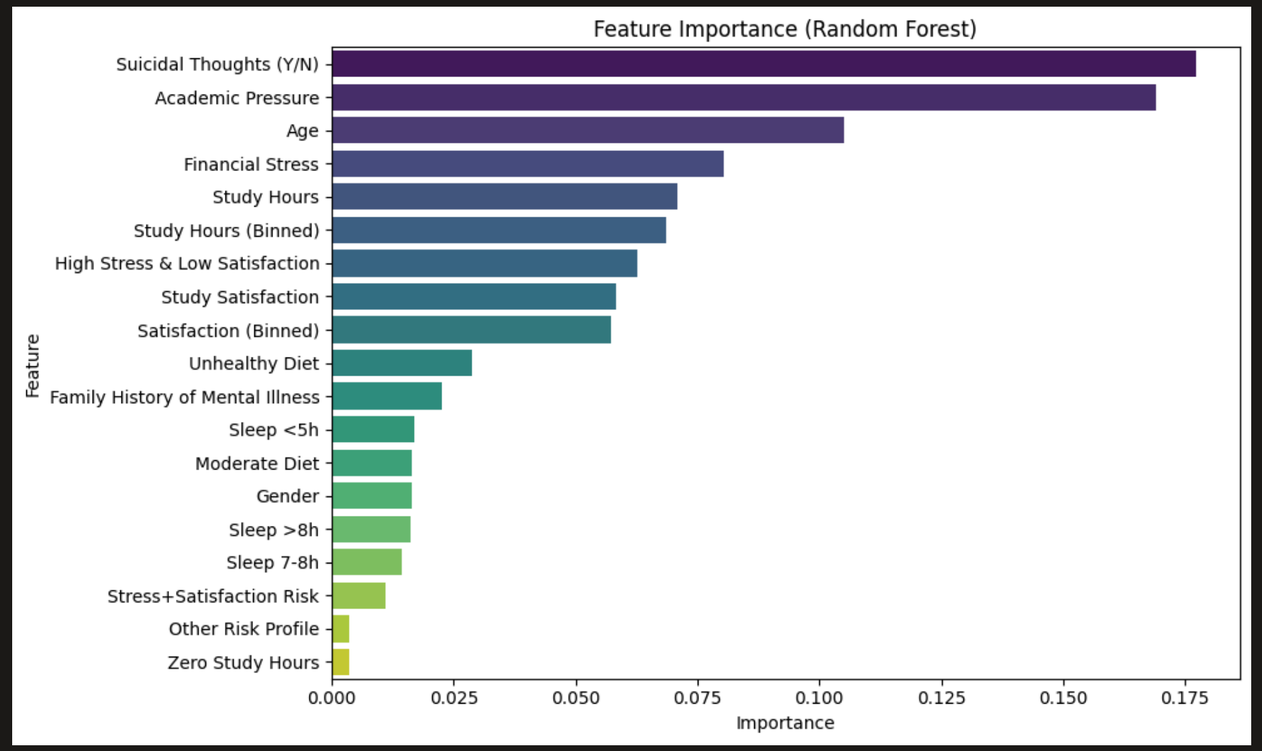
Both models achieved high accuracy, confirming that survey and behavioral variables are strong predictors of depression risk.

Both models demonstrated strong performance in identifying at-risk students, with very few misclassifications.

💡 Spotlight on Key Predictors

To understand which factors most strongly influence depression risk, we examined feature importance using both logistic regression and random forest models.





Positive values increase depression risk; negative values reduce it.

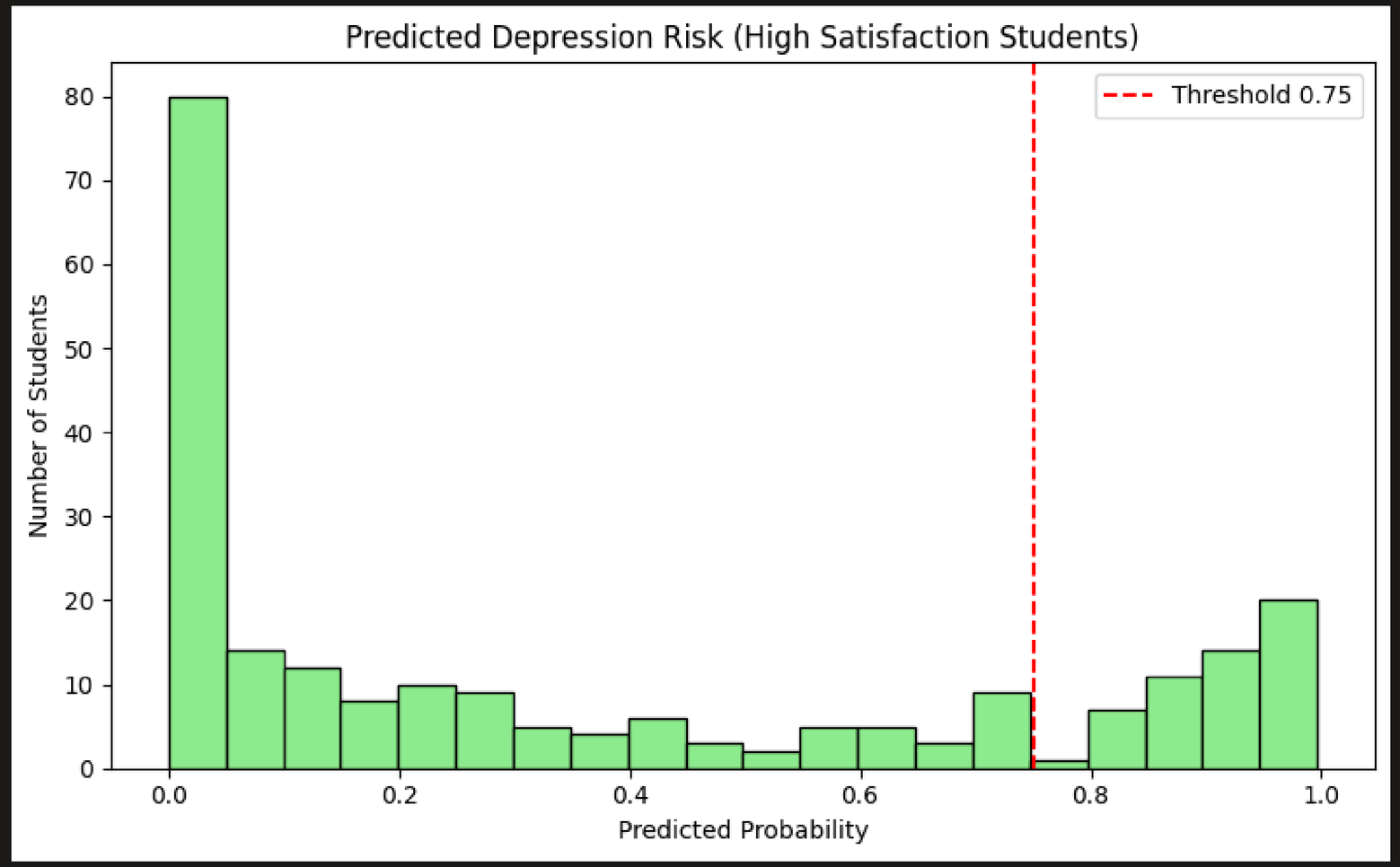
Higher bars mean the feature is more important for prediction, regardless of direction.

Academic pressure and suicidal thoughts stood out as the most consistent predictors across models.

The Mask Effect: Hidden Depression Risk Behind High Satisfaction

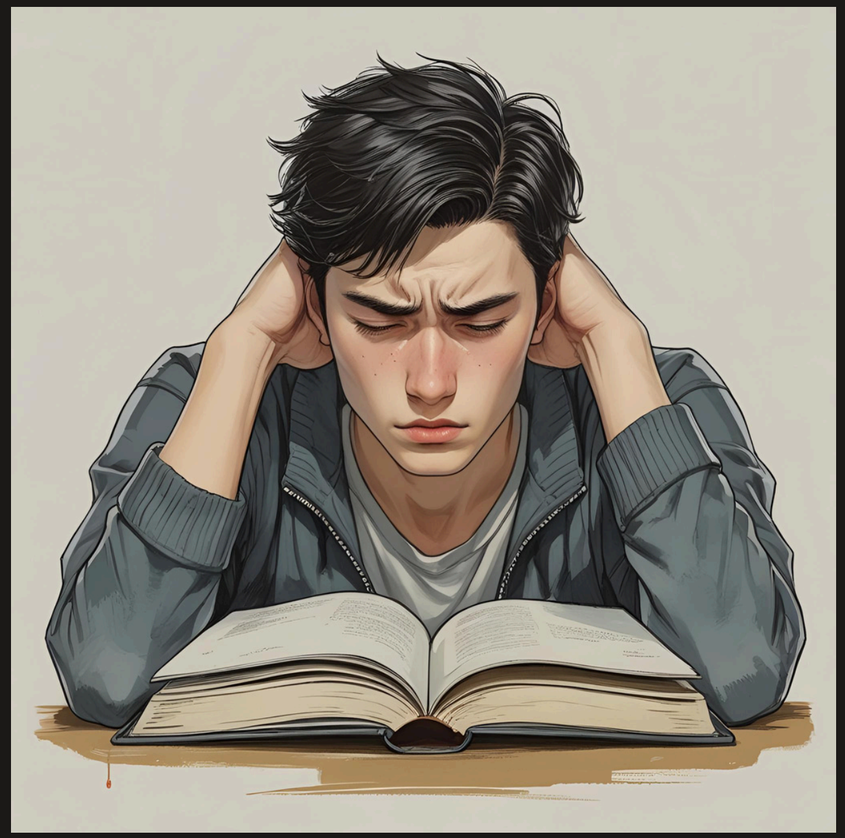
What This Shows:

- Even students who report high study satisfaction can have a high predicted probability of depression.
- Self-reported satisfaction does not always protect against mental health risks.
- This suggests some students maintain a facade of engagement while struggling internally.



Sample: Students with Study Satisfaction >0.6
Threshold: Predicted Probability >0.75 marked as High Risk

So ...Who is Most at Risk ? ⚠️



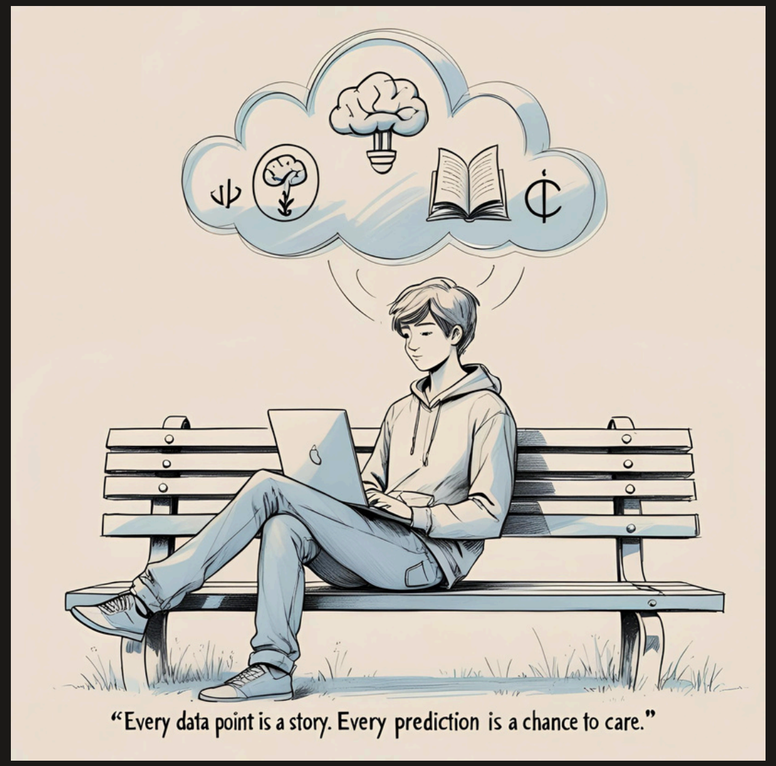
High-Risk Profile

-Age: 22
-Gender: Male
-Academic Pressure: Very High
-Study Satisfaction: Very Low
-Suicidal Thoughts: Present
-Financial Stress: High
-Study Hours: Maximum
-Diet: Unhealthy

This profile was identified by combining feature importance and predicted probabilities from our models.Students with this combination of high pressure, low satisfaction, and psychosocial stressors were consistently predicted as being at highest risk of depression.

What Did We Learn?

- Depression among students isn’t random — it’s shaped by academic, emotional, and financial factors.
- Combining surveys with online sentiment can help us spot early warning signs.
- Predictive models are powerful tools, but their real value lies in creating more supportive environments.



How This Helps ?

- Early detection of hidden depression risk
- Targeted support for students
- Complementing self-reports with predictive modeling

Thank you!